

Aditya Singh

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[LinkedIn](#) — [GitHub](#)

PROFESSIONAL SUMMARY

M.Sc. Data Science student at **IIIT Lucknow** with a strong foundation in Python, SQL, Machine Learning, NLP, and AI application development. Experienced in building end-to-end ML and NLP applications using Scikit-learn, TensorFlow, Streamlit, and Python. Familiar with LLMs, embeddings, RAG concepts, APIs, and GenAI application development. Seeking an **AI Engineer** internship or full time role to build production-oriented AI solutions and LLM-powered applications.

Technical Skills

Programming: Python, SQL, C

AI & Machine Learning: Scikit-learn, TensorFlow, Keras, NLP, Deep Learning, Machine Learning

GenAI & LLM: LLMs, Prompt Engineering, Embeddings, RAG, Vector Databases

AI Development: REST APIs, Streamlit, AI Application Development

Data & Databases: Pandas, NumPy, MySQL, PostgreSQL

Tools: Git, GitHub, Jupyter Notebook, Excel

Visualization: Power BI, Matplotlib, Seaborn

PROJECTS

- **Real-Time Pest Detection using ResNet-50 and Vision Transformer** — [GitHub](#)

Developed a hybrid deep learning framework using ResNet-50, Vision Transformer, and Custom CNN for real-time agricultural pest detection. Achieved **96%** training accuracy through fusion-based image classification and efficient feature extraction.

- **Flipkart Product Review Sentiment Analysis | NLP** — [GitHub](#) | [Live Demo](#)

Built and deployed an end-to-end NLP application to classify the sentiment of **100K+** Flipkart product reviews using TF-IDF, Logistic Regression, and Naive Bayes. Implemented text preprocessing, feature engineering, and model evaluation using Accuracy, Precision, Recall, F1-score, and Confusion Matrix. Deployed the solution as an interactive Streamlit web application for real-time sentiment prediction.

- **Bank Loan Risk Analysis | SQL** — [GitHub](#)

Analyzed 10,000+ loan records using SQL to identify high-risk customer segments and calculate default rates. Wrote 30+ SQL queries using JOINS, CTEs, GROUP BY, aggregate functions, CASE WHEN, and window functions for business analysis.

- **AI-Powered Research & Document Intelligence Assistant | RAG** — [GitHub](#) | [Live Demo](#)

Build a RAG system where a user can upload multiple research papers company reports technical PDFs and ask questions. The system retrieves relevant information and generates an answer with citations showing exactly which document/page the answer came from.

EDUCATION

- **M.Sc. Data Science (2025–2027)**

Indian Institute of Information Technology Lucknow (**IIIT Lucknow**)

CGPA: 8.0

- **B.Sc. (Hons.) Mathematics (2020–2023)**

Babu Banarasi Das University, Lucknow

CGPA: 8.69

COURSES

Database Management Systems (DBMS)

Applied Statistics

Machine Learning

Data Structures & Algorithms

Big Data Analytics

Natural Language Processing(NLP)

ACHIEVEMENTS

- INSPIRE Scholar – DST, Government of India (2020–2023)

- IIT JAM 2025 Qualified – AIR 1245